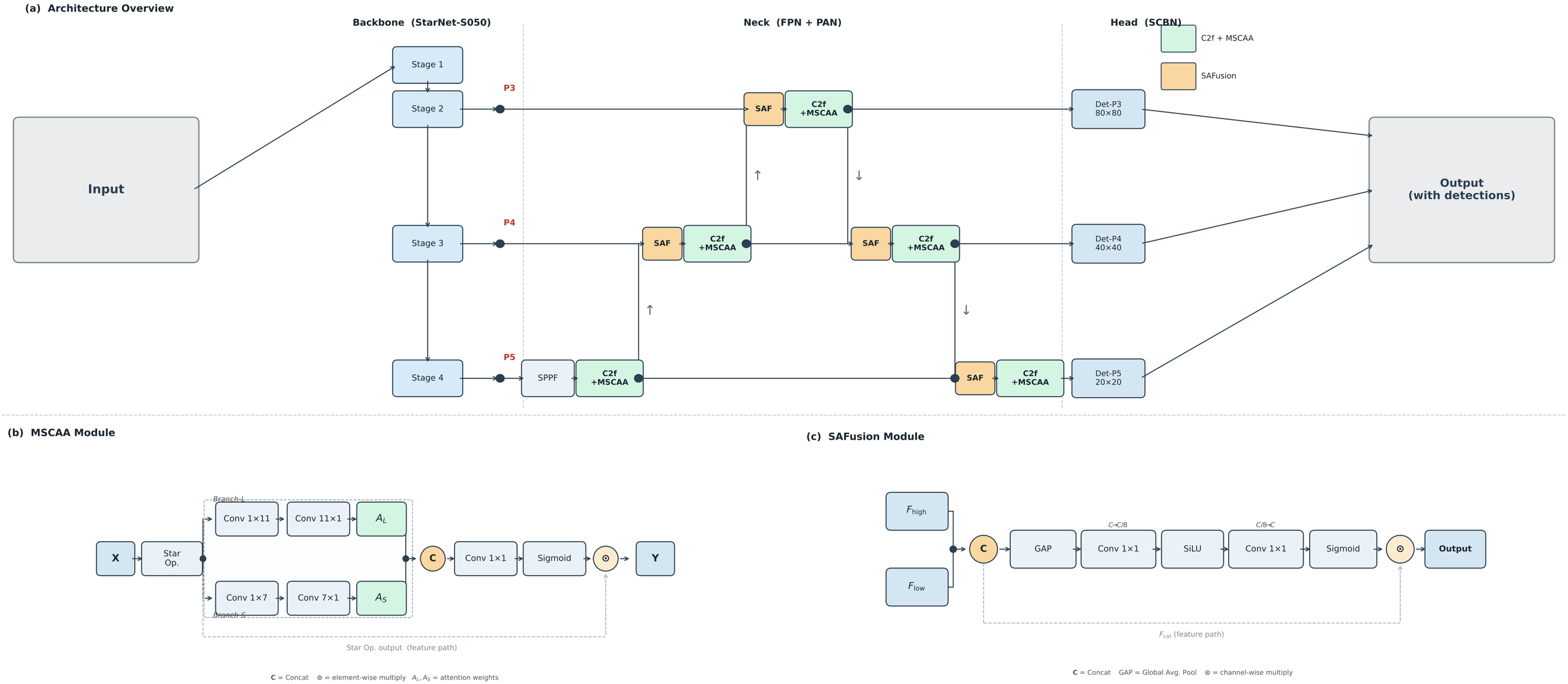
(a) Architecture Overview
Backbone (StarNet-S050)
Neck (FPN + PAN)
Head (SCBN)
C2f + MSCAA
SAFusion
Input
Stage 1
Stage 2
Stage 3
Stage 4
P3
P4
P5
SAF
C2f +MSCAA
SPPF
Det-P3 80×80
Det-P4 40×40
Det-P5 20×20
Output (with detections)
(b) MSCAA Module
X
Star Op.
Branch-L
Conv 1×11
Conv 11×1
$A_L$
Conv 1×7
Conv 7×1
$A_S$
Branch-S
C
Conv 1×1
Sigmoid
Y
Star Op. output (feature path)
C = Concat ⊙ = element-wise multiply $A_L, A_S$ = attention weights
(c) SAFusion Module
$F_{high}$
$F_{low}$
C
GAP
C→C/8
Conv 1×1
SiLU
C/8→C
Conv 1×1
Sigmoid
⊙
Output
$F_{cat}$ (feature path)
C = Concat GAP = Global Avg. Pool ⊙ = channel-wise multiply